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MyDesign Portfolio

Element A Initial Template

Element A: Presentation and Justification of the Problem

Problem Statement and Details (A1, A2, A5)

Schools don't have a reliable way to track when students go to the bathroom. Paper passes, and verbal permission is often ineffective, causing/allowing the students to spend more time outside of class, leading to disruptions in learning and teachers not knowing who is outside the class at any given time.

The main people who are involved in this problem are students and teachers, since the lack of efficient communication between them is what is causing the problem. The problem usually occurs during school hours, at any school across the world within most second and first-world countries. The issue often occurs due to insufficient communication between the teachers and students, causing the students to walk around the school, skipping class, without the teacher being able to track them.

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Problems that are connected to this problem include hallway overcrowding, loss of instructional time and learning, as well as difficulty tracking the students around the building during emergencies or class periods. These issues create safety concerns and reduce productivity within the classroom. A digital or physical prototype could solve these problems by recording student check-ins and outs, making tracking far faster and accurate. It could also be used to provide data to the school, allowing for better hallway management and student safety.

Stakeholder Concerns and Connections (A3, A4)



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- Students are a primary stakeholder because they are the ones facing the issue and causing it.
- Teachers are primary because they are the ones dealing with the issue and keeping track of students.
- Administration are a primary stakeholder due to them overseeing everything and being responsible for the teachers the students and the building.
- Parents are secondary since they don't face the issue directly, but are responsible for the student that do
- Security/Maintenance Staff are secondary(same as parents).

Problem Importance (A6)

Schools continue to face problems with student bathroom management because traditional paper pass systems are difficult to monitor and easy to misuse. When teachers rely on verbal permission or physical passes, they may lose track of which students are out of the classroom and for how long. This can contribute to hallway overcrowding, reduced instructional time, and decreased accountability. Some schools have already begun implementing electronic hall pass systems to better monitor student movement and improve school safety (Newstimes, 2025). Research on RFID and Arduino-based tracking systems also shows that digital logging systems can improve efficiency and reduce human error compared to manual tracking methods (AAR Research Index, 2024).

The problem is important because classroom time is valuable, and repeated disruptions can negatively affect learning. According to studies on RFID-based school monitoring



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systems, automated systems are more accurate and efficient than traditional manual systems because they provide real-time tracking and organized digital records (AAR Research Index, 2024). In addition, schools using digital hall pass systems have reported improvements in hallway management and student accountability (Newstimes, 2025). Developing a low-cost prototype using Arduino and MATLAB could help schools test an affordable solution before investing in more expensive commercial systems. This prototype could reduce classroom disruption, improve safety, and help teachers manage student movement more effectively.

MLA References

"Newtown High School Tries Electronic Hall-Pass System." Newstimes, 2025, www.newstimes.com/news/article/newtown-high-school-tries-electronic-hall-pass-21065148.php.

"RFID-Based Student Monitoring and Attendance Systems Using Arduino." AAR Research Index, 2024, ojs.aaresearchindex.com/index.php/aasgbcpjmra/article/view/16475.

"Arduino-Based Attendance Monitoring Systems and Efficiency Improvements." AAR Research Index, 2024, ojs.aaresearchindex.com/index.php/aasgbcpjmra/article/view/12810.



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INCLUDE a References List at the end of this document. Then delete all instructions (Green text) and the rubric. (I will subtract points from each document that has the instructions still in it when you submit the assignment in canvas.

Element A: Presentation and Justification of the Problem						
A1. Problem identification is _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	very clear and objective	clear and objective	somewhat clear and objective	somewhat clear or objective	unclear	missing (cannot be inferred from information included)
A2. Problem definition is _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	in considerable depth	in more than adequate depth	in adequate depth	in superficial depth	unclear	missing (cannot be inferred from information included)
A3. Problem elaboration is _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	well elaborated	elaborated	imprecisely or generally elaborated	minimally elaborated	unelaborated and/or is clearly subjective	not elaborated (cannot be inferred from information included)
A4. Concern of primary stakeholder groups. Justification of the problem highlights the concerns of _____.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	almost all primary stakeholder groups	many primary stakeholder groups	several primary stakeholder groups	at least a few primary stakeholders /stakeholder groups	only one or two stakeholders /stakeholder groups	no primary stakeholders /stakeholder groups
A5. Problem justification	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0

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is based on _____.	numerous, timely, credible, and various sources	timely, credible, and various sources	credible and timely sources	credible sources	partially-cre dible sources	essentially only the opinion of the researcher
A6. Problem justification	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
offers						
_____ -objective	consistently	mostly	generally	somewhat	unclear	no
detail.						